


```
root@sysadmin:/home/sysadmin/devops9prom
root@sysadmin:/home/sysadmin/devops9prom# helm install promgra ./promgra-0.0.1.tgz
Error: INSTALLATION FAILED: Kubernetes cluster unreachable: get "https://192.168.49.2:8443/version": dial tcp 192.168.49.2:8443: connect: no route to host
root@sysadmin:/home/sysadmin/devops9prom# swapoff -a
minikube start --driver=docker --force
@ minikube v1.38.1 on Ubuntu 24.04
! minikube skips various validations when --force is supplied; this may lead to unexpected behavior
! Using the docker driver based on existing profile
! The "docker" driver should not be used with root privileges. If you wish to continue as root, use --force.
! If you are running minikube within a VM, consider using --driver=none:
! https://minikube.sigs.k8s.io/docs/reference/drivers/none/
! Tip: To remove this root owned cluster, run: sudo minikube delete

@ The requested memory allocation of 3072MiB does not leave room for system overhead (total system memory: 3867MiB). You may face stability issues.
! Suggestion: Start minikube with less memory allocated: "minikube start --memory=3072mb"

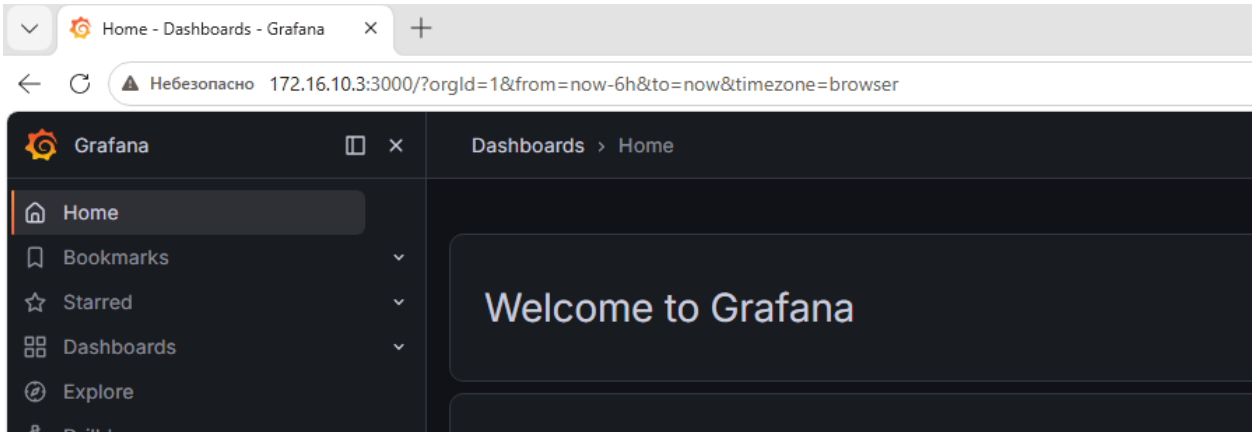
@ Starting "minikube" primary control-plane node in "minikube" cluster
@ Pulling base image v0.0.50 ...
@ Restarting existing docker container for "minikube" ...
@ Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
@ Verifying Kubernetes components...
! Using image gcr.io/k8s-minikube/storage-provisioner:v5
@ Enabled addons: default-storageclass, storage-provisioner
@ Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
root@sysadmin:/home/sysadmin/devops9prom# helm install promgra ./promgra-0.0.1.tgz
NAME: promgra
LAST DEPLOYED: Fri Jul 3 09:05:24 2026
NAMESPACE: default
STATUS: deployed
REVISIONS: 1
TEST SUITE: None
root@sysadmin:/home/sysadmin/devops9prom# kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
blackbox 0/1 1 0 18s
grafana 0/1 1 0 18s
prometheus 0/1 1 0 18s
root@sysadmin:/home/sysadmin/devops9prom# kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
blackbox ClusterIP 10.104.67.60 <none> 9115/TCP 28s
grafana LoadBalancer 10.96.232.34 172.16.10.3 3000:32231/TCP 28s
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 8h
prometheus ClusterIP 10.108.72.223 <none> 9090/TCP 28s
root@sysadmin:/home/sysadmin/devops9prom#
```

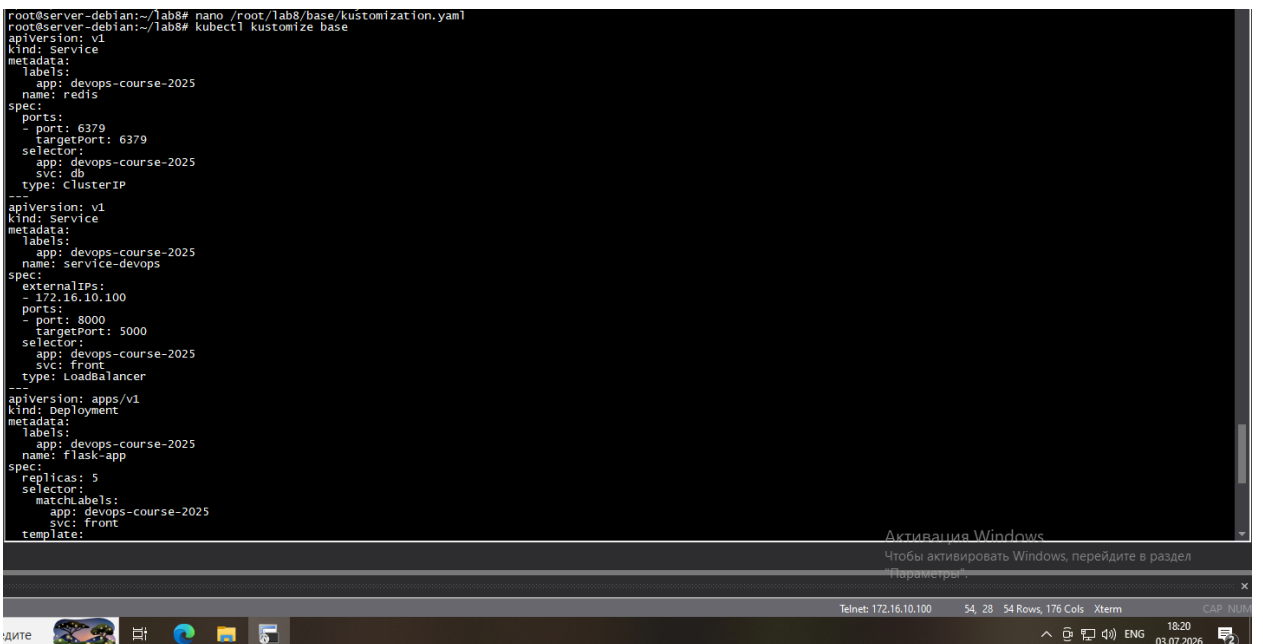
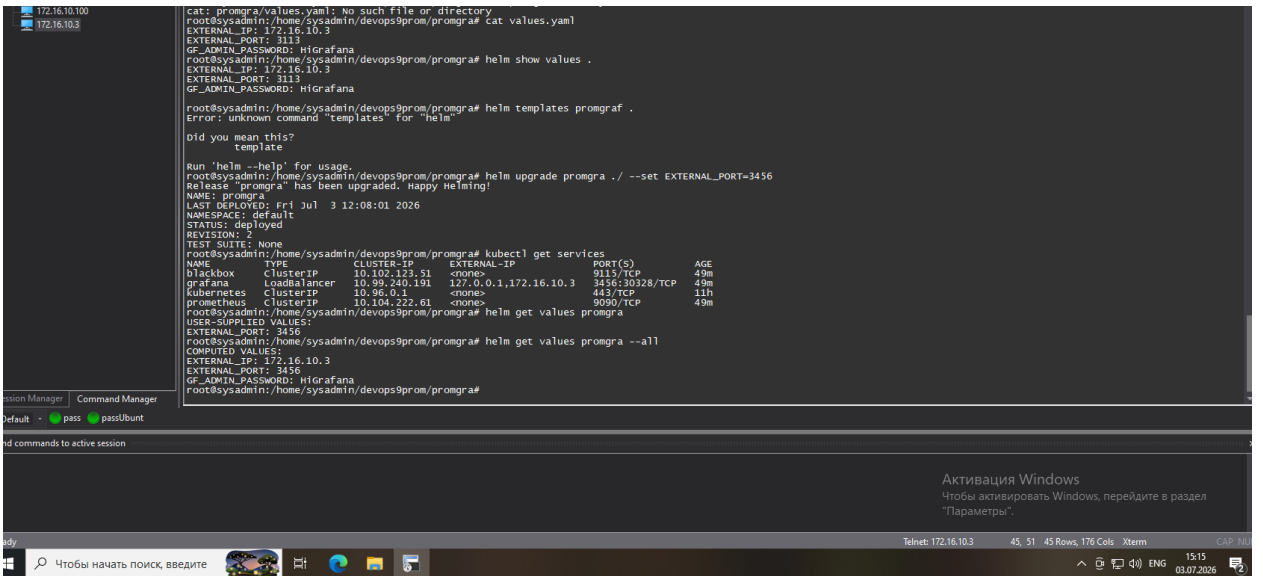
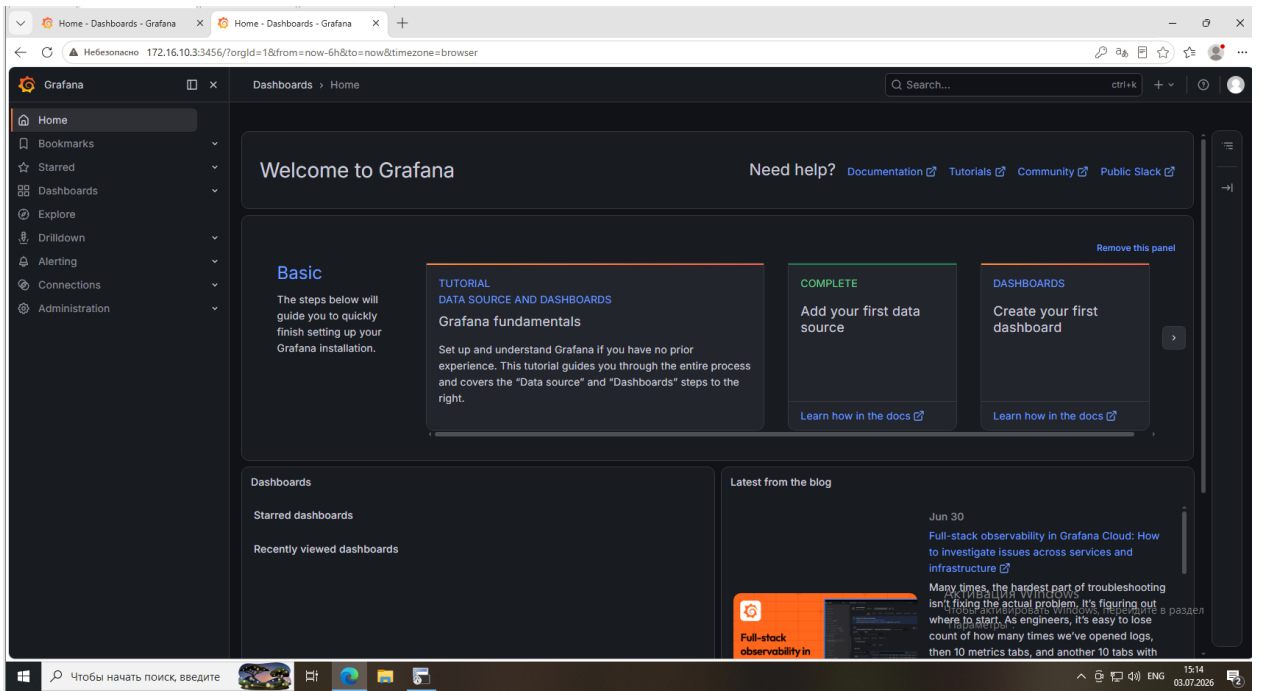
```
root@sysadmin:/home/sysadmin/devops9prom
root@sysadmin:/home/sysadmin/devops9prom# minikube status
minikube
type: Control plane
host: Stopped
kubelet: Stopped
apiserver: Stopped
kubeconfig: Stopped
root@sysadmin:/home/sysadmin/devops9prom# docker ps | grep minikube
root@sysadmin:/home/sysadmin/devops9prom# minikube start --driver=docker --force
@ minikube v1.38.1 on Ubuntu 24.04
! minikube skips various validations when --force is supplied; this may lead to unexpected behavior
! Using the docker driver based on existing profile
! The "docker" driver should not be used with root privileges. If you wish to continue as root, use --force.
! If you are running minikube within a VM, consider using --driver=none:
! https://minikube.sigs.k8s.io/docs/reference/drivers/none/
! Tip: To remove this root owned cluster, run: sudo minikube delete

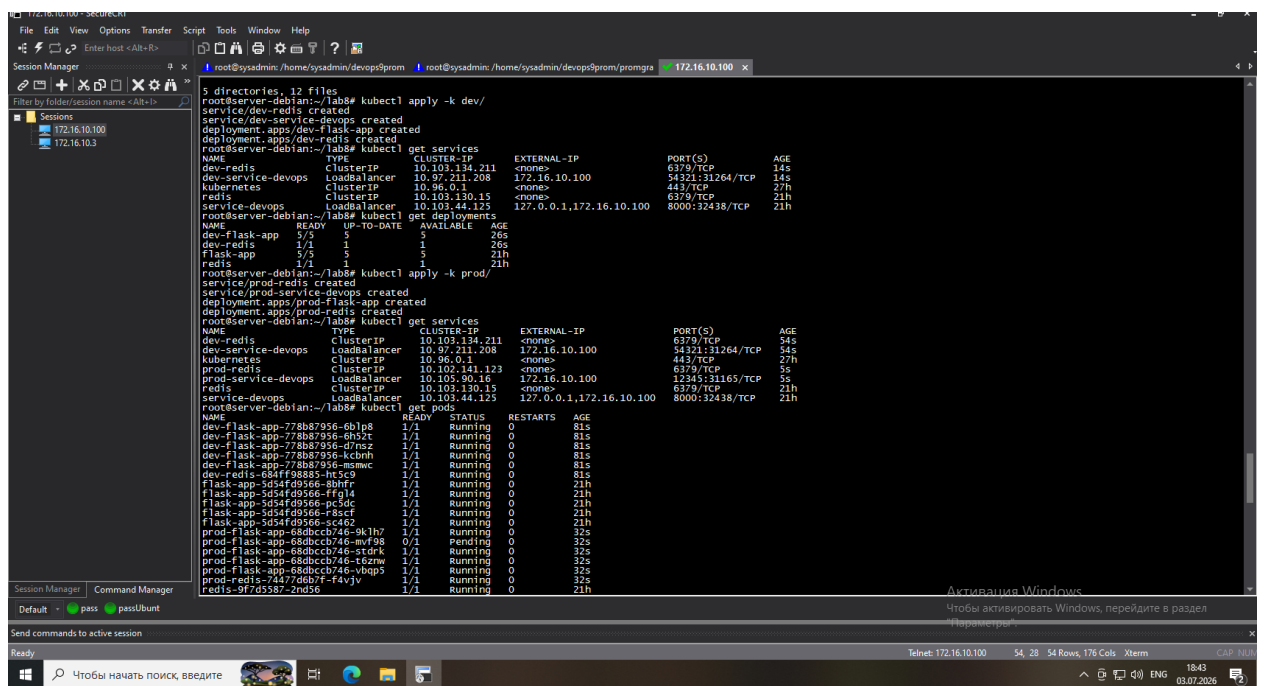
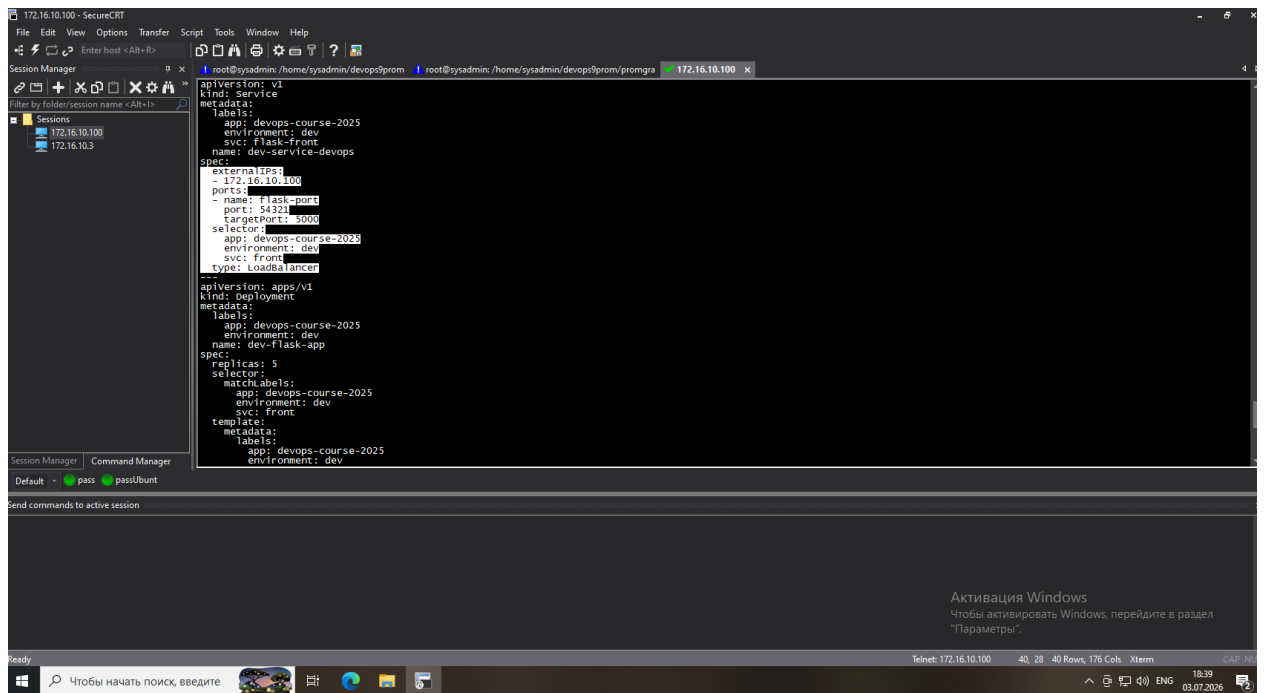
@ The requested memory allocation of 3072MiB does not leave room for system overhead (total system memory: 3868MiB). You may face stability issues.
! Suggestion: Start minikube with less memory allocated: "minikube start --memory=3072mb"

@ Starting "minikube" primary control-plane node in "minikube" cluster
@ Pulling base image v0.0.50 ...
@ Restarting existing docker container for "minikube" ...
! Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
! To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
@ Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
@ Verifying Kubernetes components...
! Using image gcr.io/k8s-minikube/storage-provisioner:v5
@ Enabled addons: default-storageclass, storage-provisioner
@ Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
root@sysadmin:/home/sysadmin/devops9prom# minikube tunnel --bind-address 172.16.10.3
Tunnel successfully started

! NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...
@ Starting tunnel for service grafana.
```







```
root@server-debian:~/lab8# kubectl apply -k dev/
service/dev-redis unchanged
service/dev-service-devops unchanged
deployment.apps/dev-flask-app configured
deployment.apps/dev-redis unchanged
root@server-debian:~/lab8# kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
dev-flask-app-778b87956-kcbnh       1/1      Running   0           52m
dev-flask-app-778b87956-msmwc       1/1      Running   0           52m
dev-redis-684ff98885-ht5c9          1/1      Running   0           52m
flask-app-5d54fd9566-8bhfr          1/1      Running   0           22h
flask-app-5d54fd9566-ffgl4          1/1      Running   0           22h
flask-app-5d54fd9566-pc5dc          1/1      Running   0           22h
flask-app-5d54fd9566-r8scf          1/1      Running   0           22h
flask-app-5d54fd9566-sc462          1/1      Running   0           22h
prod-flask-app-68dbccb746-9klh7     1/1      Running   0           51m
prod-flask-app-68dbccb746-mvf98     1/1      Running   0           51m
prod-flask-app-68dbccb746-stdrk     1/1      Running   0           51m
prod-flask-app-68dbccb746-t6znw     1/1      Running   0           51m
prod-flask-app-68dbccb746-vbqp5     1/1      Running   0           51m
prod-redis-74477d6b7f-f4vjv        1/1      Running   0           51m
redis-9f7d5587-2nd56               1/1      Running   0           22h
```

172.16.10.100 (1) - SecureCRT

File Edit View Options Transfer Script Tools Window Help

Enter host <Alt+R>

Session Manager

Filter by folder/session name <Alt+S>

Sessions

172.16.10.100

172.16.10.3

namePrefix: dev-

labels:

- includeSelectors: true

pairs:

environment: dev

patches:

- path: service-patch.yaml

target:

- labelSelector: 'svc=flask-front'

- path: deployment-patch.yaml

root@server-debian:~/lab8# nano prod/kustomization.yaml

root@server-debian:~/lab8# kubectl apply -k prod

service/prod-redis unchanged

service/prod-service-devops unchanged

deployment.apps/prod-flask-app configured

deployment.apps/prod-redis unchanged

deployment.apps/prod-flask-app restarted

root@server-debian:~/lab8# curl 172.16.10.100:12345

curl: (56) Recv failure: соединение разорвано другой стороной

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 916 times. My name is: prod-flask-app-6f45db7b4-bh5gl My env: PRODUCTION

root@server-debian:~/lab8# curl 172.16.10.100:54321

Hello world VERSION 5! I have been seen 921 times. My name is: dev-flask-app-f57bd98f9-vg4k My env: DEVELOPMENT

root@server-debian:~/lab8# curl 172.16.10.100:54321

Hello world VERSION 5! I have been seen 931 times. My name is: dev-flask-app-f57bd98f9-gl1pk8 My env: DEVELOPMENT

root@server-debian:~/lab8# curl 172.16.10.100:54321

Hello world VERSION 5! I have been seen 932 times. My name is: dev-flask-app-f57bd98f9-gl1pk8 My env: DEVELOPMENT

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 935 times. My name is: prod-flask-app-f57bd98f9-gl1pk8 My env: PRODUCTION

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 937 times. My name is: prod-flask-app-58549967fb-5458n My env: PRODUCTION

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 939 times. My name is: dev-flask-app-f57bd98f9-vg4k My env: DEVELOPMENT

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 942 times. My name is: prod-flask-app-58549967fb-h6xdf My env: PRODUCTION

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 943 times. My name is: prod-flask-app-58549967fb-wr8ch My env: PRODUCTION

root@server-debian:~/lab8# curl 172.16.10.100:12345

Hello world VERSION 5! I have been seen 945 times. My name is: prod-flask-app-58549967fb-gkh8n My env: PRODUCTION

root@server-debian:~/lab8#

Session Manager

Command Manager

Default

pass

passUbuntu

Send commands to active session

Ready

Чтобы начать поиск, введите

Телнет: 172.16.10.100 50, 28 50 Rows, 176 Cols Xterm

2016 03.07.2026

Активация Windows

Чтобы активировать Windows, перейдите в раздел "Параметры".

```
root@server-debian:~/lab8# kubectl get services
NAME                                TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
dev-redis                           ClusterIP    10.103.134.211   <none>            6379/TCP         98m
dev-service-devops                  ClusterIP    10.97.211.208    127.0.0.1,172.16.10.100 34321:31264/TCP  98m
kubernetes                          ClusterIP    10.96.0.1        <none>            443/TCP          28h
prod-redis                          ClusterIP    10.102.141.123   <none>            6379/TCP         97m
prod-service-devops                 LoadBalancer 10.103.90.16     127.0.0.1,172.16.10.100 12345:31165/TCP  97m
redis                               ClusterIP    10.103.130.15    <none>            6379/TCP         23h
service-devops                      LoadBalancer 10.103.44.125    127.0.0.1,172.16.10.100 8000:32438/TCP   23h
root@server-debian:~/lab8#
```

Активация Windows

Чтобы активировать Windows, перейдите в раздел "Параметры".

Телнет: 172.16.10.100 50, 28 50 Rows, 176 Cols Xterm

2018 03.07.2026